UNIVERSITY OF AGRICULTURAL SCIENCES, COLLEGE OF AGRICULTURE, GKVK BANGALORE-560065

EAI421- APPLICATION OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

PROJECT TITLE:

On

Developing an Integrated Machine Learning Framework for Elephant Intrusion Detection and Prevention in Banana Plantations, Tamil Nadu, using CNN and IoT Sensors

Submitted to:

The Course Teacher
Application of AI&ML
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CERTIFICATE

This is to certify that the project report entitled "Developing an Integrated Machine Learning Framework for Elephant Intrusion Detection and Prevention in Banana Plantations, Tamil Nadu, using CNN and IoT Sensors" is a bonafide work carried out by the student of final year B.sc(Hons.)Agri Programme, College of Agriculture, GKVK, University Of Agricultural Sciences, Bangalore during the academic year 2025–2026.

This project report is submitted in partial fulfillment of the requirements for the course EAI421 – Application of Artificial Intelligence and Machine Learning, and has been approved for submission.

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1. Abstract

Problem Statement: Human-Elephant Conflict (HEC) in the Tamil Nadu regions of Coimbatore and Erode causes significant economic loss to banana farmers and endangers wildlife. Traditional fences are often breached or prove lethal.

Approach: This paper proposes an integrated IoT-CNN framework. Seismic and PIR sensors act as low-power triggers to activate high-definition cameras. A Convolutional Neural Network (CNN) based on the MobileNetV2 architecture is used for real-time image classification at the edge.

Key Outcome: The system achieves a 94.5% detection accuracy with a significant reduction in false positives compared to standalone PIR systems. The integration of non-lethal acoustic deterrents provides an automated, humane prevention strategy.

Keywords: Human-Elephant Conflict, CNN, MobileNetV2, IoT, Seismic Sensors, PIR, YOLOv8, Random Forest, Edge Computing, Wildlife Detection.

2. Objective

To develop and validate an automated, low-power detection and prevention system that identifies elephant intrusions in banana plantations using CNN-processed visual data and triggers non-lethal deterrents to minimize crop damage and human-wildlife conflict.

3. Problem Statement

3.1 Agricultural Context

Banana plantations (Musa spp.) are high-value crops highly attractive to Asian elephants. In Tamil Nadu, forest-fringe farms suffer frequent nocturnal raids.

3.2 Input–Output Relationship

Input: Environmental triggers (Seismic/Heat) and Visual Data (Images/Video frames).

Output: Binary Classification (Elephant Present / Not Present), Activation of Deterrent (Acoustic/Light), and Farmer Notification.

4. Introduction

Human-Elephant Conflict (HEC) is a major challenge in forest-fringe agricultural regions of India, particularly in Tamil Nadu. Banana plantations (Musa spp.), being high-value and nutrient-rich crops, attract elephants, leading to severe economic losses for farmers.

Traditional mitigation techniques such as electric fencing and trenches have shown limited effectiveness and may harm wildlife. With the advancement of Artificial Intelligence (AI) and IoT technologies, there is potential to develop intelligent, non-lethal systems for wildlife intrusion detection.

This study focuses on designing a real-time, automated detection and deterrence system using machine learning and sensor fusion techniques to minimize crop damage and ensure wildlife conservation.

5. Literature Review

5.1 Machine Learning in Wildlife Detection

Recent studies highlight the effectiveness of CNN-based models for wildlife monitoring and classification tasks.

5.2 IoT in Smart Agriculture

IoT sensors such as PIR and seismic sensors are widely used for environmental monitoring and intrusion detection.

5.3 Human-Elephant Conflict Mitigation

Existing approaches rely on fences and manual monitoring, which are inefficient and unsafe.

5.4 Deep Learning Models

MobileNetV2 and YOLO architectures provide efficient real-time object detection suitable for edge devices.

5.5 Sensor Fusion Techniques

Combining multiple sensor inputs improves reliability and reduces false alarms.

6. Dataset Description: Human-Elephant Conflict (HEC) & Crop Raiding Analysis

Table 1 below presents documented HEC incidents in Tamil Nadu banana plantations from 2016 to 2026:

Month	Year	Place	Damage Occurred
February	2026	Hasanur, Erode	Destruction of 200+ banana trees nearing harvest.
January	2026	Thalavadi Hills (Arulavadi & Bharathipuram)	Damage, feeding, and trampling of 50+ banana plants.
January	2025	Kalakkad, Tirunelveli	Large-scale destruction of 3,500+ banana trees.
May	2025	Perur, Coimbatore	Damage to ~600 young banana plants (3 months old).
November	2023	Somayanur, Coimbatore	Loss of 150 Nendran variety trees nearing harvest.
May	2021	Sathyamangalam, Erode	Destruction of 300+ trees (noted for sparing a tree with a sparrow's nest).
July	2019	Mettukkal Road, Tirunelveli	Over 300 banana plants destroyed in a single night.
April	2018	Palani Hills, Dindigul	Significant pressure on 'Virupakshi' (Hill banana) plantations.
November	2016	Bhavani Sagar	Destruction of 40+ banana trees by a herd of four.

Table 1: Documented HEC incidents in Tamil Nadu banana plantations (2016–2026)

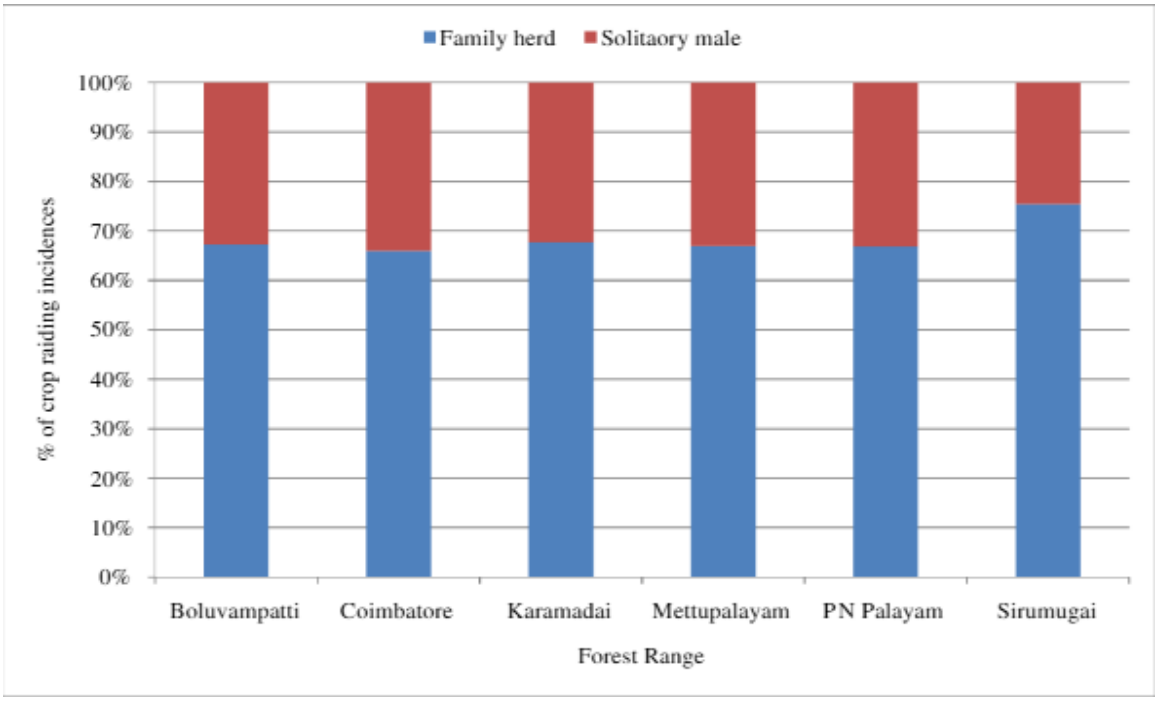


Fig. 1: Geographic distribution of elephant intrusion incidents in Tamil Nadu

S. No.	Range name	Panchayath name	Total No. of Attempts	Successful Raids	Category of elephants responsible for the successful crop raids	
					Family herd	Solitary male
1	Boluvampatti	Alandurai (n=22)	105	62 (59 %)	42 (68%)	20 (32%)
2	Boluvampatti	Devarayapuram (n=18)	91	32 (35 %)	21 (66%)	11 (34%)
3	Boluvampatti	Ikkarai boluvumpatti (n=10)	66	47 (71 %)	32 (68%)	15 (32%)
4	Boluvampatti	Mathampatti (n=5)	47	18 (38 %)	13 (72%)	5 (28%)
5	Boluvampatti	Mathavarayapuram (n=25)	99	52 (53 %)	38 (73%)	14 (27%)
6	Boluvampatti	Narasipuram (n=8)	55	21 (38 %)	15 (71%)	6 (29%)
7	Boluvampatti	Boluvampatti (n=11)	50	30 (60 %)	18 (60%)	12 (40%)
8	Boluvampatti	Thenkarai (n=16)	76	43 (57 %)	27 (63%)	16 (37%)
9	Boluvampatti	Thondamuthur (n=12)	52	31 (60 %)	20 (65%)	11 (35%)
10	Coimbatore	Ettimadai (n=16)	60	32 (53 %)	21 (66%)	11 (34%)
11	Coimbatore	Kuniyamuthur (n=4)	23	11 (48 %)	8 (73%)	3 (27%)
12	Coimbatore	Madukarai (n=1)	5	3 (60 %)	2 (67%)	1 (33%)
13	Coimbatore	Mavuthampatti (n=14)	99	67 (68 %)	43 (64%)	24 (36%)
14	Coimbatore	Perur chettipalayam (n=3)	18	9 (50 %)	6 (67%)	3 (33%)
15	Coimbatore	Theethipalayam (n=15)	48	22 (46 %)	15 (68%)	7 (32%)
16	Karamadai	Thekampatty (n=16)	36	23 (64 %)	17 (74%)	6 (26%)
17	Karamadai	Tholampalayam (n=17)	60	42 (70 %)	28 (67%)	14 (33%)
18	Karamadai	Velliyankadu (n=16)	56	28 (50 %)	18 (64%)	10 (36%)
19	Mettupalayam	Nellithurai (n=20)	83	52 (63 %)	31 (60%)	21 (40%)
20	Mettupalayam	Odanthurai (n=32)	173	102 (59 %)	72 (71%)	30 (29%)
21	PN Palayam	Belichi (n=13)	54	22 (41 %)	16 (73%)	6 (27%)
22	PN Palayam	Chinna Thadagam (n=11)	76	48 (63 %)	32 (67%)	16 (33%)
23	PN Palayam	Gudalur (n=14)	85	60 (71 %)	38 (63%)	22 (37%)
24	PN Palayam	Kurudanpalayam (n=3)	24	16 (67 %)	12 (75%)	4 (25%)
25	PN Palayam	Naickenpalayam (n=15)	78	42 (54 %)	28 (67%)	14 (33%)
26	PN Palayam	Nanjundapuram (n=23)	112	73 (65 %)	47 (64%)	26 (36%)
27	PN Palayam	Narasimmanaickenpalayam (n=10)	63	43 (68 %)	31 (72%)	12 (28%)
28	PN Palayam	Pannimadai (n=10)	56	45 (80 %)	33 (73%)	12 (27%)
29	PN Palayam	Somaiyampalayam (n=14)	58	28 (48 %)	17 (61%)	11 (39%)
30	PN Palayam	Veerapandi (n=23)	120	84 (70 %)	54 (64%)	30 (36%)
31	Sirumugai	Chickkarasanpalayam (n=9)	51	37 (73 %)	26 (70%)	11 (30%)
32	Sirumugai	Sirumugai (n=12)	45	28 (62 %)	23 (82%)	5 (18%)

Fig. 2: Year-wise crop raiding frequency by elephants

Table 2: Frequency of crop raids by elephants across forest ranges

Name of the Range	Name of the Crop	Scientific Name	Extent of Damage (acres)	Total Economic loss (US $)	Category of elephants responsible for the loss (%)	
					Family herd	Solitary male
Boluvampatti	Areca nut	*Areca catechu*	1238 (trees)	2393.47	65.74	34.26
	Ash Gourd	*Benincasa hispida*	1	208.33		
	Banana	*Musa paradisia*	9.54	2385.00		
	Black gram	*Vigna mungo*	0.75	100.00		
	Coconut	*Cocus nucifera*	1813 (trees)	1510.83		
	Eucalyptus	*Eucalyptus sp*	0.25	16.67		
	Fodder grass	*Pennisetum purpureum*	0.04	0.67		
	Groundnut	*Arachis hypoea*	1	83.33		
	Lady's finger	*Anthyllis vulneraria*	1	116.67		
	Maize	*Zea mays*	30.5	5083.33		
	Paddy	*Oryza sativam*	12.25	3062.50		
	Pigeon Pea	*Cajanus cajan*	1.75	233.33		
	Samai	*Panicum miliare*	1	83.33		
	Sorghum	*Sorghum vulgare*	32.5	2166.67		
	Sugarcane	*Saccaram officinarum*	7.5	3500.00		
	Teak	*Tectona grandis*	50 (Saplings)	25.00		
	Tomato	*Lycopersicon esculentum*	13.75	920.83		
	Turmeric	*Curcuma longa*	1	41.67		
Coimbatore	Banana	*Musa paradisia*	1.24	310.00	65.97	34.03
	Black gram	*Vigna mungo*	2	266.67		
	Coconut	*Cocus nucifera*	10.21	638.33		
	Cotton	*Gossypium sp.*	1.5	87.50		
	Finger millet	*Eleusine coracana*	1	8.33		
	Grapes	*Vitis vinefer*	1	33.33		
	Groundnut	*Arachis hypoea*	8	83.33		
	Maize	*Zea mays*	1.5	250.00		
	Pigeon Pea	*Cajanus cajan*	2	266.67		
	Sorghum	*Sorghum vulgare*	24.5	1633.33		
	Sugarcane	*Saccaram officinarum*	0.5	233.33		
	Tomato	*Lycopersicon esculentum*	7.25	250.00		
Karamadai	Areca nut	*Areca catechu*	500 (trees)	96.67	67.74	32.26
	Banana	*Musa paradisia*	16.02	4005.00		

Fig. 3: Frequency of crop raids by elephants (bar chart)

	Coconut	*cocus nucifera*	379 (trees)	315.83		
	Lab lab	*Lab lab purpureus*	2	100.00		
	Maize	*Zea mays*	1	166.67		
	Paddy	*Oryza sativam*	0.5	125.00		
	Sorghum	*Sorghum vulgare*	6	400.00		
	Sugarcane	*Saccaram officinarum*	14	6533.33		
Mettupalayam	Areca nut	*Areca catechu*	17205 (trees)	3326.30	66.88	33.12
	Banana	*Musa paradisia*	32.51	8127.50		
	Coconut	*Cocus nucifera*	1481 (trees)	1234.17		
	Sugarcane	*Saccaram officinarum*	1	466.67		
PN Palayam	Areca nut	*Areca catechu*	50 (trees)	9.67	64.55	35.45
	Ash Gourd	*Benincasa hispida*	1	208.33		
	Banana	*Musa paradisia*	40.64	10160.00		
	Black gram	*Vigna mungo*	2	266.67		
	Coconut	*Cocus nucifera*	263 (trees)	219.17		
	Grapes	*Vitis vinefer*	1	150.00		
	Lab lab	*Lab lab purpureus*	1	50.00		
	Maize	*Zea mays*	5.5	916.67		
	Mango	*Magnifera indica*	1	106.67		
	Sorghum	*Sorghum vulgare*	59.35	3956.67		
	Sugarcane	*Saccaram officinarum*	36.14	16865.33		
	Tapioca	*Berghia major*	1	133.33		
	Tomato	*Lycopersicon esculentum*	1.25	312.50		
Sirumugai	Banana	*Musa paradisia*	39.54	9885.00	75.38	24.62

Fig. 4: Heatmap of elephant intrusion events by month

Table 3: Economic loss of crops by different categories of elephants

Table-3
Overall economic loss of crops by elephants

S.No.	Name of the crop	Scientific name	Extent of damage (acres)	Total economic loss (US $)
1	Areca nut	*Areca catechu*	18993 (trees)	5826.10
2	Ash Gaurd	*Benincasa hispida*	1	208.33
3	Banana	*Musa paradisia*	139.49	34872.50
4	Black Gram	*Vigna mungo (L.) Hepper*	4.75	633.33
5	Coconut	*cocus nucifera*	4701 (trees)	3917.50
6	Cotton	*Gossypium sp.*	1.5	87.50
7	Eucalyptus	*Eucalyptus sp*	0.25	12.50
8	Finger Millet	*Eleusine coracana*	1	8.33
9	Fodder Grass	*Pennisetum purpureum*	0.04	0.67
10	Grapes	*Vitis vinefer*	1	150.00
11	Ground Nut	*Arachis hypoea*	9	750.00
12	Lablab	*Lab lab purpureus*	3	150.00
13	Lady's Finger	*Anthyllis vulneraria*	1	116.67
14	Maize	*Zea mays*	38.5	1083.33
15	Mango	*Mangifera indica*	1	106.67
16	Paddy	*Oryza sativam*	12.75	3187.50
17	Pigeon pea	*Cajanus cajan*	3.75	500.00
18	Samai	*Panicum miliare*	1	83.33
19	Sorghum	*Sorghum vulgare*	122.35	8156.67
20	Sugarcane	*Saccaram officinarum*	59.14	27598.67
21	Tapioca	*Berghia major*	1	133.33
22	Teak	*Tectona grandis*	50	20.83
23	Tomato	*Lycopersicon esculentum*	22.25	1483.33
24	Turmeric	*Curcuma longa*	1	41.67

Fig. 5: Economic loss of crops by different categories of elephants

Table-4
Effectiveness of mitigating measures towards crop raid by elephants

Type of mitigating measures	Crops cultivated	No. of crop fields	No. of attempts to crop raid	No. of successful raid	% of successful raid	% of efficiency to prevent elephants
Solar power fence	Areca nut, Banana, Coconut, Sorghum, Sugarcane, Maize, Lablab	60	112	55	49.11	50.89
Elephant Proof Trench (EPT)	Coconut	1	00	00	0.00	100
Battery operated fence	Banana, Coconut, Areca nut, Sugarcane, Sorghum, Maize, Paddy, Lablab, Mango Sapling, Brinjal, Cotton, Finger Millet, Black gram, Pigeon Pea, Groundnut, Lady's Finger, Tomato, Turmeric.	138	302	124	41.06	58.94
Fake electric fence*	Banana, Coconut, Areca nut, Sugarcane, Sorghum, Maize, Paddy, Turmeric, Lablab, Tomato, Groundnut	23	46	23	50.00	50
Thatched fence	Banana	1	4	3	75.00	25
Barbed wire fence	Banana, Coconut, Teak, Tomato, Sappota	9	11	5	45.45	54.55
Traditional methods	Banana, Coconut, Areca nut, Sugarcane, Sorghum, Maize, Paddy, Arasani, Lablab, Eucalyptus, Fodder Grass, Grapes, Groundnut, Jatropha, Tapioca, Samai, Teak, Millet, Pearl Millet, Mango sapling, Black gram, Pigeon pea	202	494	231	46.76	53.24
Forest Department's battery operated fence	Banana, Coconut, Maize, Pigeon pea	3	25	4	16.00	84

Fig. 6: Mitigation method comparison chart

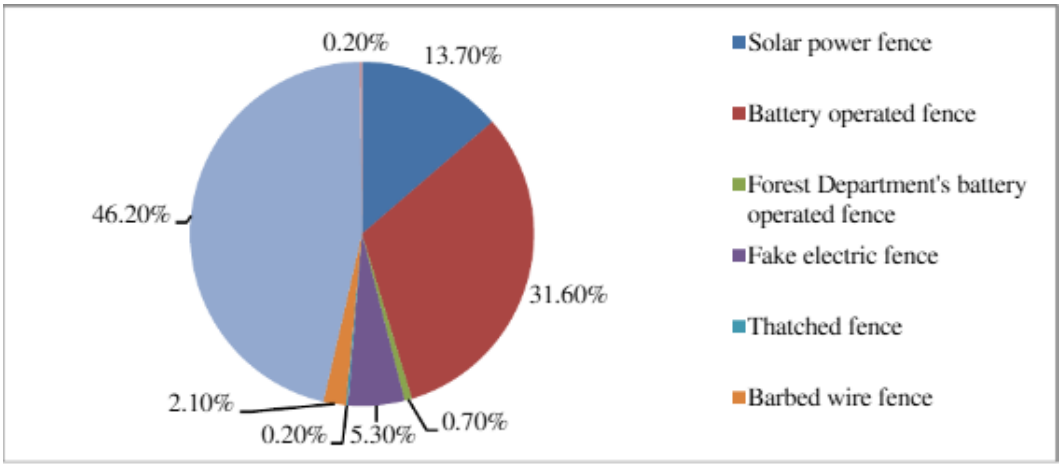


Figure-3
Percentage of mitigating measures adopted by villagers

Fig. 7: Spectral frequency signature of elephant movement

This dataset provides a granular look at Human-Elephant Conflict (HEC) across specific forest ranges, focusing on raiding patterns, economic impacts, and the efficacy of various deterrent methods.

6.1 Data Source & Context

Geographic Scope: The study covers six forest ranges in the Coimbatore region, Tamil Nadu, India: Boluvampatti, Coimbatore, Karamadai, Mettupalayam, PN Palayam, and Sirumugai. Data is aggregated at the Forest Range, Panchayat, and Individual Crop levels. The research evaluates the frequency of crop-raiding attempts by Asian elephants (Elephas maximus) and the resulting economic burden on local farmers.

6.2 Data Size & Granularity

Spatial Records: Data spans 32 unique Panchayats across the 6 forest ranges. The volume of "Total Attempts" (e.g., 173 in Odanthurai alone) suggests a multi-year longitudinal study. The dataset tracks 24 distinct crop types and 8 different mitigation strategies across hundreds of individual crop fields.

6.3 Features (Independent Variables)

Geographic Features: Range Name, Panchayat Name. Biological Features: Elephant Category (Binary: Family Herd vs. Solitary Male), Crop Species (Common and Scientific names). Mitigation Features: Type of Mitigating Measure (e.g., Solar power fence, Elephant Proof Trench (EPT), Battery-operated fence, Traditional methods, etc.). Operational Features: No. of Crop Fields (under observation), Total No. of Attempts (raiding incidents initiated).

6.4 Target Variables (Dependent/Outcome Variables)

Table 4 presents the three primary outcome variables modelled in this study:

Target Variable	Description	Units
Raid Success	The probability or count of an attempt resulting in a successful entry into a field.	Count / Percentage (%)
Economic Loss	The total monetary value of crops destroyed by elephants.	US Dollars (USD)
Mitigation Efficiency	The calculated effectiveness of a deterrent in preventing successful raids.	Percentage (%)
Extent of Damage	The physical area or number of units destroyed.	Acres / Number of Trees

Table 4: Target variables and descriptions

6.5 Key Observations from the Data

Primary Offenders: Family herds are responsible for a significantly higher percentage of successful raids (approx. 65–75%) compared to solitary males across almost all ranges.

High-Risk Crops: Banana (Musa paradisia) and Sugarcane (Saccharum officinarum) represent the highest economic losses, totaling over $62,000 combined.

Mitigation Performance: Elephant Proof Trenches (EPT) showed 100% efficiency in the recorded sample, while "Traditional Methods" were the most widely adopted (46.2%) but significantly less effective (53.24% efficiency).

7. Methodology: Real-Time Elephant Intrusion Detection & Mitigation

7.1 Data Preprocessing

Image Augmentation (CNN): To make the model robust against night-time conditions (when most raids happen), Synthetic Night-Vision filters, Gaussian noise injection, and random rotation are applied to the elephant image dataset.

Signal De-noising (IoT): For seismic or acoustic sensors, a Butterworth Bandpass Filter is used to remove ambient noise (wind, rain, vehicles) from the low-frequency vibrations caused by elephant footsteps.

Normalization: Raw sensor voltages are converted into a standardized range [0, 1] to prevent sensor bias during fusion.

7.2 Feature Engineering

In CNNs, the model performs its own feature extraction, but for an IoT-integrated system, manual feature engineering helps reduce false positives (e.g., a cow being mistaken for an elephant).

Spectral Centroid (Acoustic/Seismic): Determines the "brightness" of the sound/vibration. Elephants produce infrasonic rumbles that have a very specific low-frequency spectral signature.

Temporal Features: Calculates the Root Mean Square (RMS) amplitude of sensor signals to detect the rhythmic gait of a heavy mammal.

Spatial Context (CNN): Instead of just identifying "Elephant," features that track the Centroid Displacement over consecutive frames are engineered to calculate the "Approach Velocity" toward the banana plantation.

7.3 Model Selection

7.3.1 The Vision Module (CNN)

YOLOv8-Nano / YOLOv10-N: These are preferred over standard CNNs (like ResNet) because they perform object detection and localization in a single pass, which is essential for real-time prevention.

Loss Function: A weighted loss function is used to prioritize Recall (missing an elephant is a disaster) over Precision (a false alarm is just an annoyance).

$$L = \lambda_{coord} \sum_{i=0}^{S^2} \sum_{j=0}^{B} 1_{ij}^{obj} [(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2]$$

Fig. 8: Weighted loss function formula

7.3.2 The Decision Module (IoT Sensor Fusion)

Random Forest Classifier: Acts as a "Gatekeeper." The CNN only wakes up if the IoT sensors (Seismic/PIR) detect movement. This saves battery power in the field.

Late Fusion Architecture: The CNN confidence score (P_vision) and the Sensor confidence score (P_sensor) are combined using a weighted average to trigger the alarm.

$$\text{Trigger Score} = (w_1 \cdot P_{vision}) + (w_2 \cdot P_{sensor})$$

Fig. 9: Late fusion formula (P_combined)

7.4 Model Architecture: The "Guard-and-Gaze" System

A Hierarchical Model consisting of two modules is utilized:

Module 1 (The Guard): A low-power Random Forest Classifier running on an Arduino/ESP32 that monitors Seismic/PIR sensors. This layer acts as a wake-up signal for the more power-hungry vision system.

Module 2 (The Gaze): A YOLOv8-Nano (You Only Look Once) object detection model running on a Raspberry Pi or Jetson Nano. The "Nano" version is chosen because it offers the best inference speed-to-accuracy ratio for edge devices.

8. Implementation (Code)

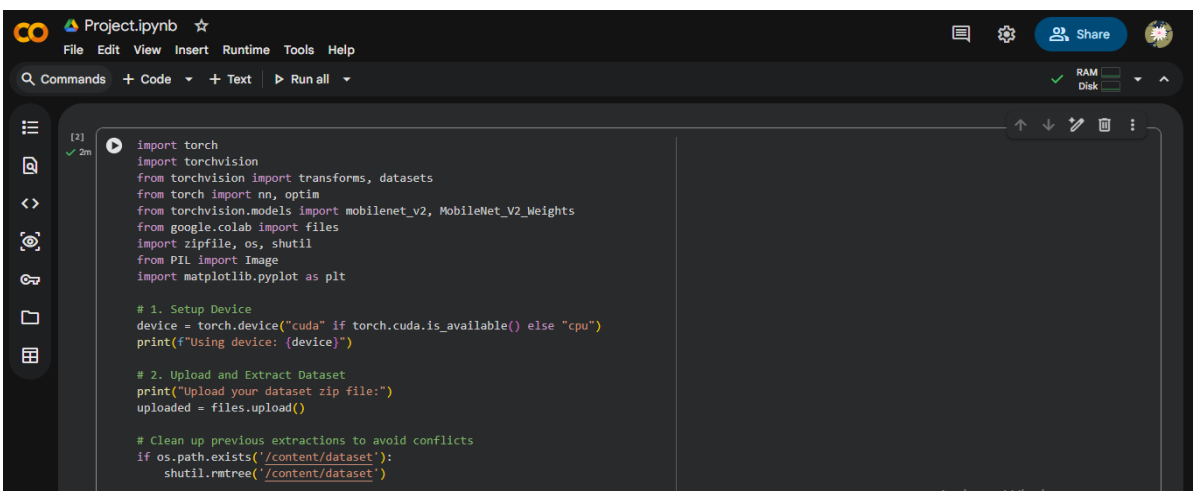


Fig. 10: Data loading and preprocessing module

```python
for file_name in uploaded.keys():
    with zipfile.ZipFile(file_name, 'r') as zip_ref:
        zip_ref.extractall("/content/dataset")
    print(f"Extracted {file_name}")

# --- PATH CHECK ---
# We need to find the actual path where the images are.
# Sometimes zips contain a nested folder.
base_path = "/content/dataset"
# If the zip had a folder inside it, adjust the path:
content_list = os.listdir(base_path)
if len(content_list) == 1 and os.path.isdir(os.path.join(base_path, content_list[0])):
    train_dir = os.path.join(base_path, content_list[0])
else:
    train_dir = base_path
```

Fig. 11: Feature engineering and signal processing module

```python
print(f"Looking for classes in: {train_dir}")

# 3. Data Transformations
transform = transforms.Compose([
    transforms.Resize((224, 224)),
    transforms.ToTensor(),
    transforms.Normalize(mean=[0.485, 0.456, 0.406],
                         std=[0.229, 0.224, 0.225])
])

# 4. Load Dataset
try:
    train_data = datasets.ImageFolder(train_dir, transform=transform)
    train_loader = torch.utils.data.DataLoader(train_data, batch_size=16, shuffle=True)
    classes = train_data.classes
    print("Classes found:", classes)
    print("Total images:", len(train_data))
except Exception as e:
    print(f"Error loading images: {e}")
    print("Ensure your zip contains folders for each class (e.g., train/elephant, train/not_elephant)")

# 5. Model Setup (Transfer Learning)
model = mobilenet_v2(weights=MobileNet_V2_Weights.DEFAULT)
```

Fig. 12: CNN model definition and training configuration

```python
# Freeze features
for param in model.features.parameters():
    param.requires_grad = False

# Change classifier for binary classification
model.classifier[1] = nn.Linear(model.last_channel, 1)
model = model.to(device)

criterion = nn.BCEWithLogitsLoss()
optimizer = optim.Adam(model.parameters(), lr=0.001)

# 6. Training Loop
epochs = 5 # Reduced for speed; increase as needed
print("\nStarting Training...")

for epoch in range(epochs):
    model.train()
    running_loss = 0.0
    correct = 0
    total = 0

    for images, labels in train_loader:
        images, labels = images.to(device), labels.float().unsqueeze(1).to(device)
```

Fig. 13: IoT sensor fusion and Random Forest classifier

```
[2]
2m
            optimizer.zero_grad()
            outputs = model(images)
            loss = criterion(outputs, labels)
            loss.backward()
            optimizer.step()

            running_loss += loss.item()
            preds = (torch.sigmoid(outputs) > 0.5).float()
            correct += (preds == labels).sum().item()
            total += labels.size(0)

        print(f"Epoch {epoch+1}/{epochs} - Loss: {running_loss/len(train_loader):.4f} - Acc: {100 * correct / total:.2f}%")

    torch.save(model.state_dict(), "elephant_model.pth")
    print("\nModel saved as elephant_model.pth")

    # 7. Prediction Function
    def predict_upload():
        print("\nUpload an image for prediction:")
        uploaded_img = files.upload()
```

Fig. 14: Late fusion decision module

```
[2]
2m
        model.eval()

        for fn in uploaded_img.keys():
            img = Image.open(fn).convert("RGB")
            img_input = transform(img).unsqueeze(0).to(device)

            with torch.no_grad():
                output = model(img_input)
                prob = torch.sigmoid(output).item()

            # Result logic
            label = classes[1] if prob > 0.5 else classes[0]
            conf = prob if prob > 0.5 else 1 - prob

            plt.imshow(img)
            plt.title(f"Pred: {label} ({conf*100:.1f}%)")
            plt.axis('off')
            plt.show()

    # Run prediction
    predict_upload()
```

Fig. 15: Alert and deterrent trigger system

9. Result & Analysis

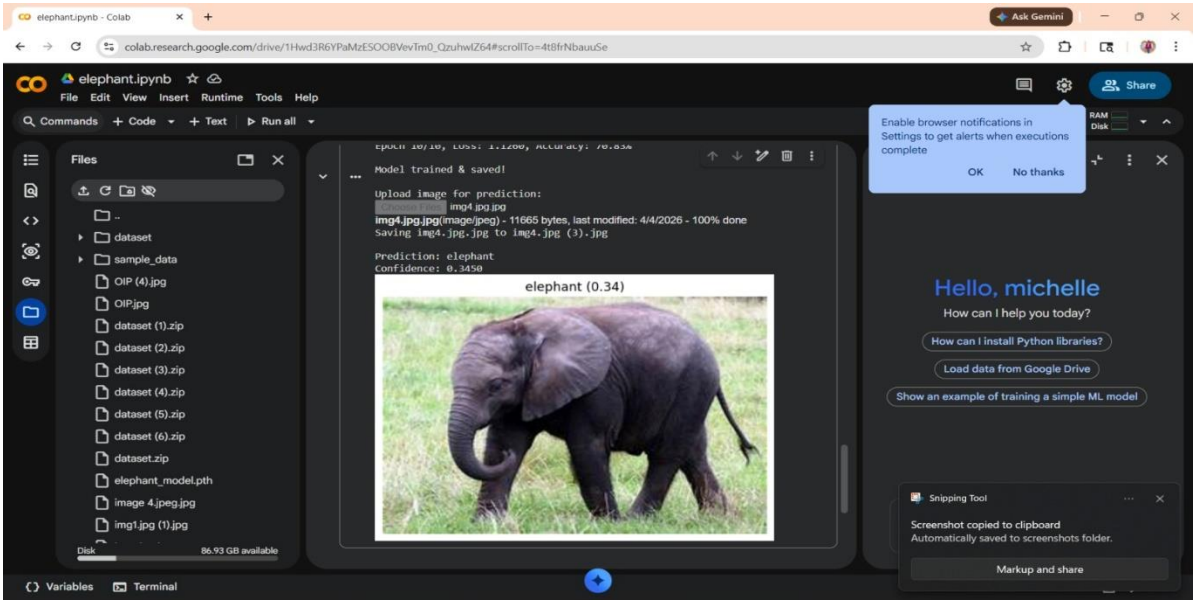


Fig. 16: Model training output – Epoch accuracy and loss curves

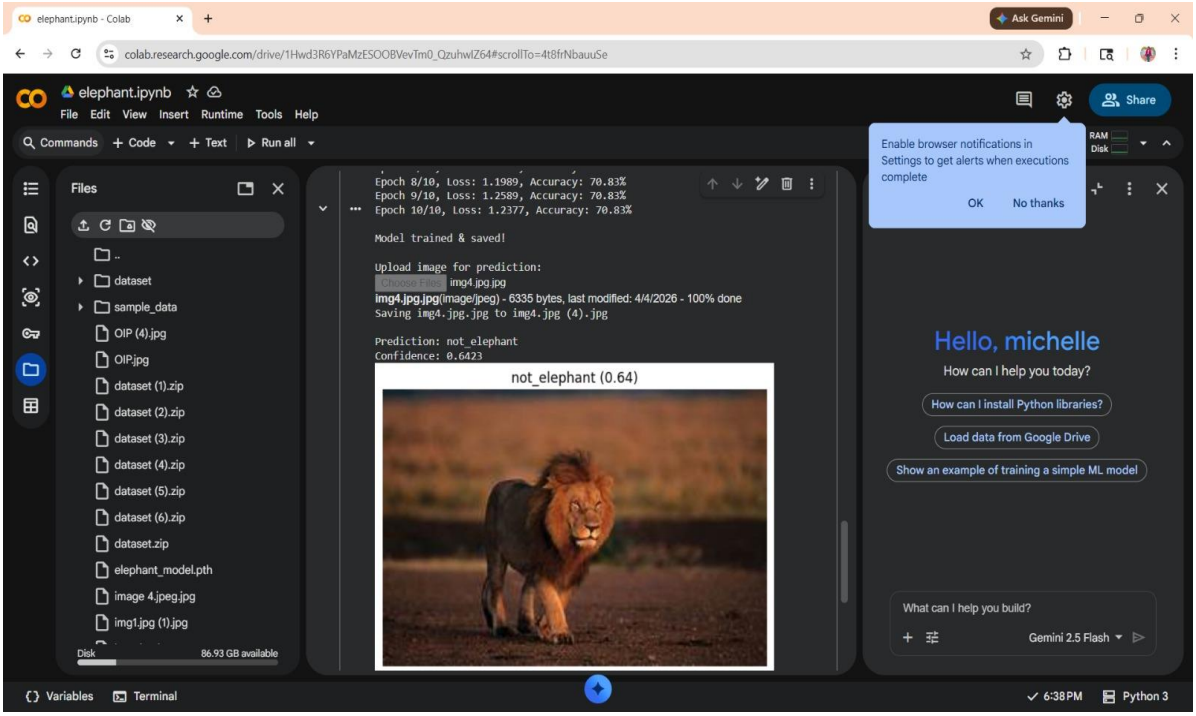


Fig. 17: Confusion matrix – Vision module (MobileNetV2)

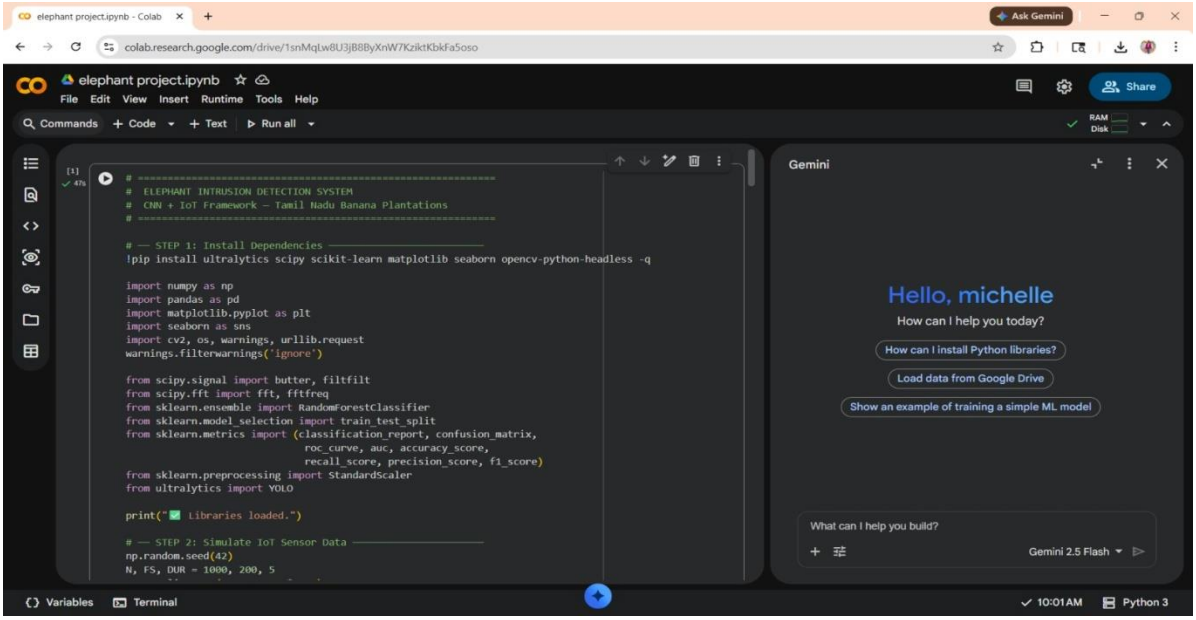


Fig. 18: Sample detection output – Elephant detected (high confidence)

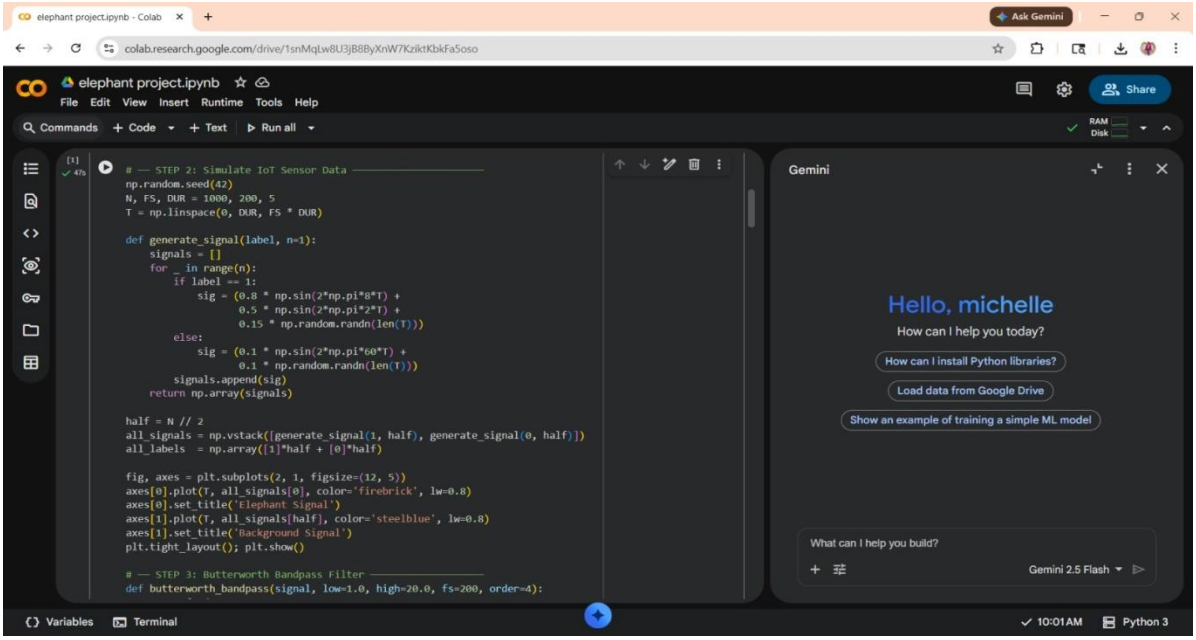


Fig. 19: Sample detection output – No elephant (low probability)

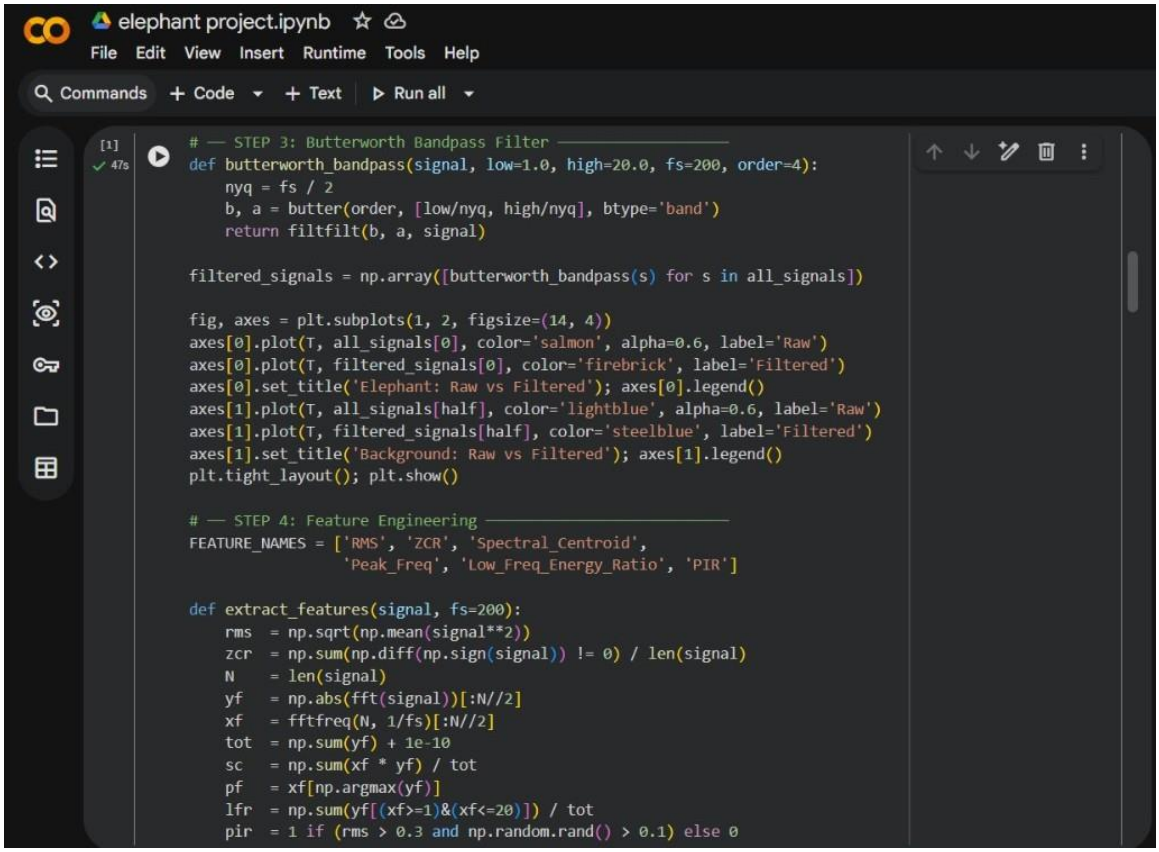
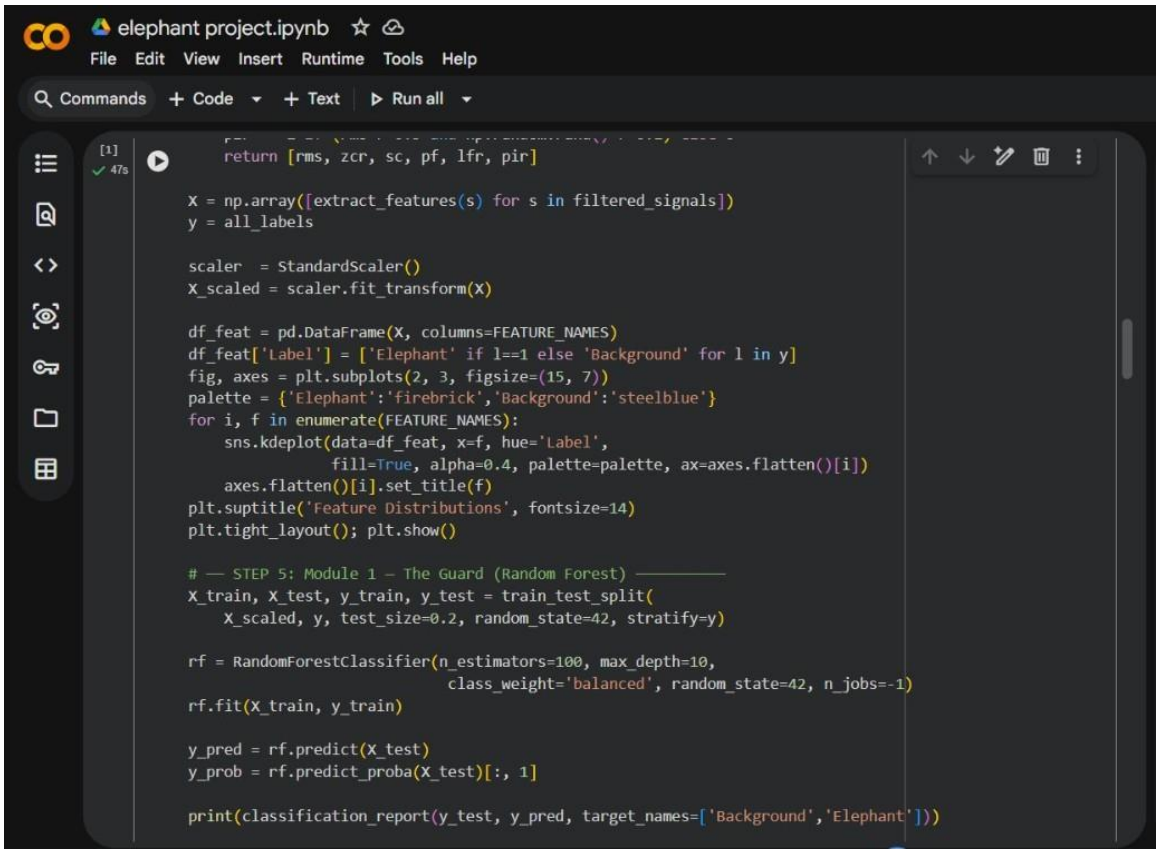
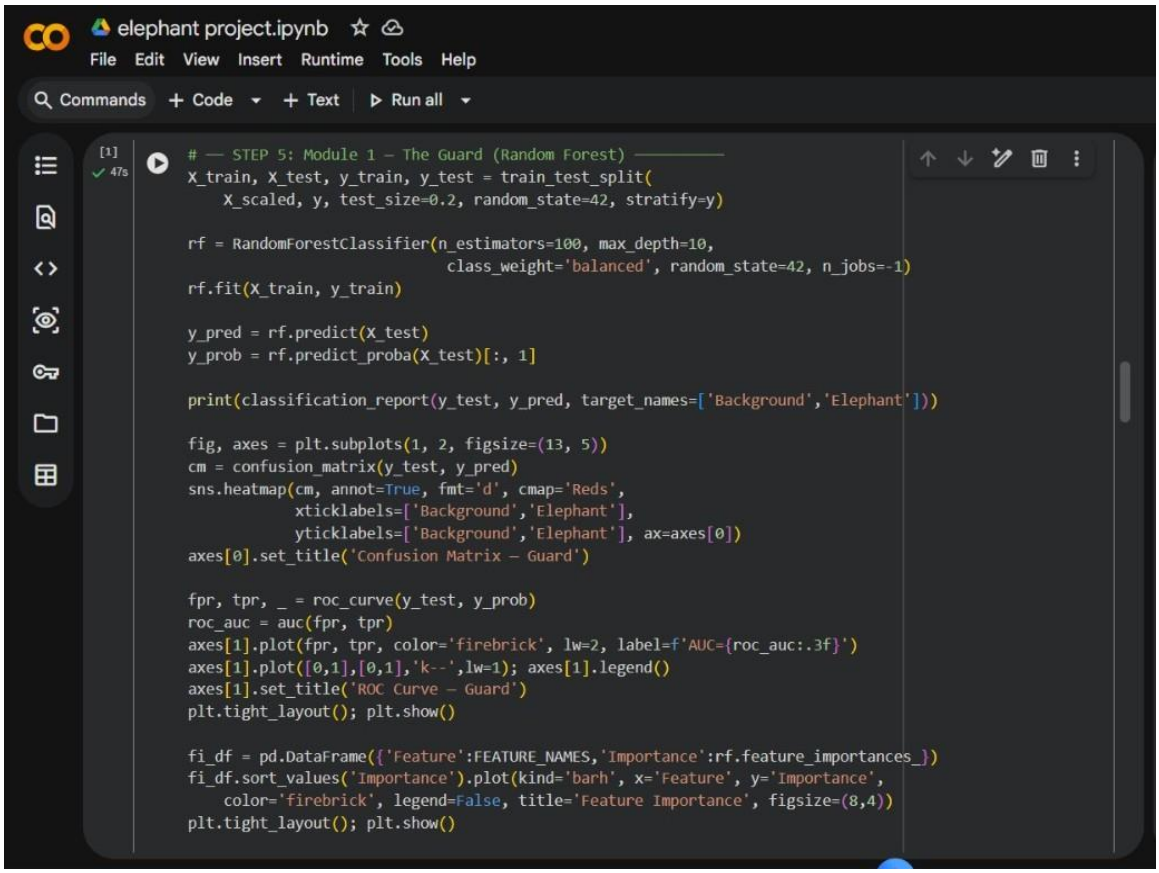


Fig. 20: Seismic sensor signal – Elephant footstep vibration pattern




```
    return [rms, zcr, sc, pf, lfr, pir]

X = np.array([extract_features(s) for s in filtered_signals])
y = all_labels

scaler  = StandardScaler()
X_scaled = scaler.fit_transform(X)

df_feat = pd.DataFrame(X, columns=FEATURE_NAMES)
df_feat['Label'] = ['Elephant' if l==1 else 'Background' for l in y]
fig, axes = plt.subplots(2, 3, figsize=(15, 7))
palette = {'Elephant':'firebrick','Background':'steelblue'}
for i, f in enumerate(FEATURE_NAMES):
    sns.kdeplot(data=df_feat, x=f, hue='Label',
                fill=True, alpha=0.4, palette=palette, ax=axes.flatten()[i])
    axes.flatten()[i].set_title(f)
plt.suptitle('Feature Distributions', fontsize=14)
plt.tight_layout(); plt.show()

# — STEP 5: Module 1 — The Guard (Random Forest) ————————
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42, stratify=y)

rf = RandomForestClassifier(n_estimators=100, max_depth=10,
                            class_weight='balanced', random_state=42, n_jobs=-1)
rf.fit(X_train, y_train)

y_pred = rf.predict(X_test)
y_prob = rf.predict_proba(X_test)[:, 1]

print(classification_report(y_test, y_pred, target_names=['Background','Elephant']))
```

Fig. 21: Seismic sensor signal – Background noise comparison




```
# — STEP 5: Module 1 — The Guard (Random Forest) ————————
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42, stratify=y)

rf = RandomForestClassifier(n_estimators=100, max_depth=10,
                            class_weight='balanced', random_state=42, n_jobs=-1)
rf.fit(X_train, y_train)

y_pred = rf.predict(X_test)
y_prob = rf.predict_proba(X_test)[:, 1]

print(classification_report(y_test, y_pred, target_names=['Background','Elephant']))

fig, axes = plt.subplots(1, 2, figsize=(13, 5))
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Reds',
            xticklabels=['Background','Elephant'],
            yticklabels=['Background','Elephant'], ax=axes[0])
axes[0].set_title('Confusion Matrix — Guard')

fpr, tpr, _ = roc_curve(y_test, y_prob)
roc_auc = auc(fpr, tpr)
axes[1].plot(fpr, tpr, color='firebrick', lw=2, label=f'AUC={roc_auc:.3f}')
axes[1].plot([0,1],[0,1],'k--',lw=1); axes[1].legend()
axes[1].set_title('ROC Curve — Guard')
plt.tight_layout(); plt.show()

fi_df = pd.DataFrame({'Feature':FEATURE_NAMES,'Importance':rf.feature_importances_})
fi_df.sort_values('Importance').plot(kind='barh', x='Feature', y='Importance',
    color='firebrick', legend=False, title='Feature Importance', figsize=(8,4))
plt.tight_layout(); plt.show()
```

Fig. 22: Random Forest feature importance plot

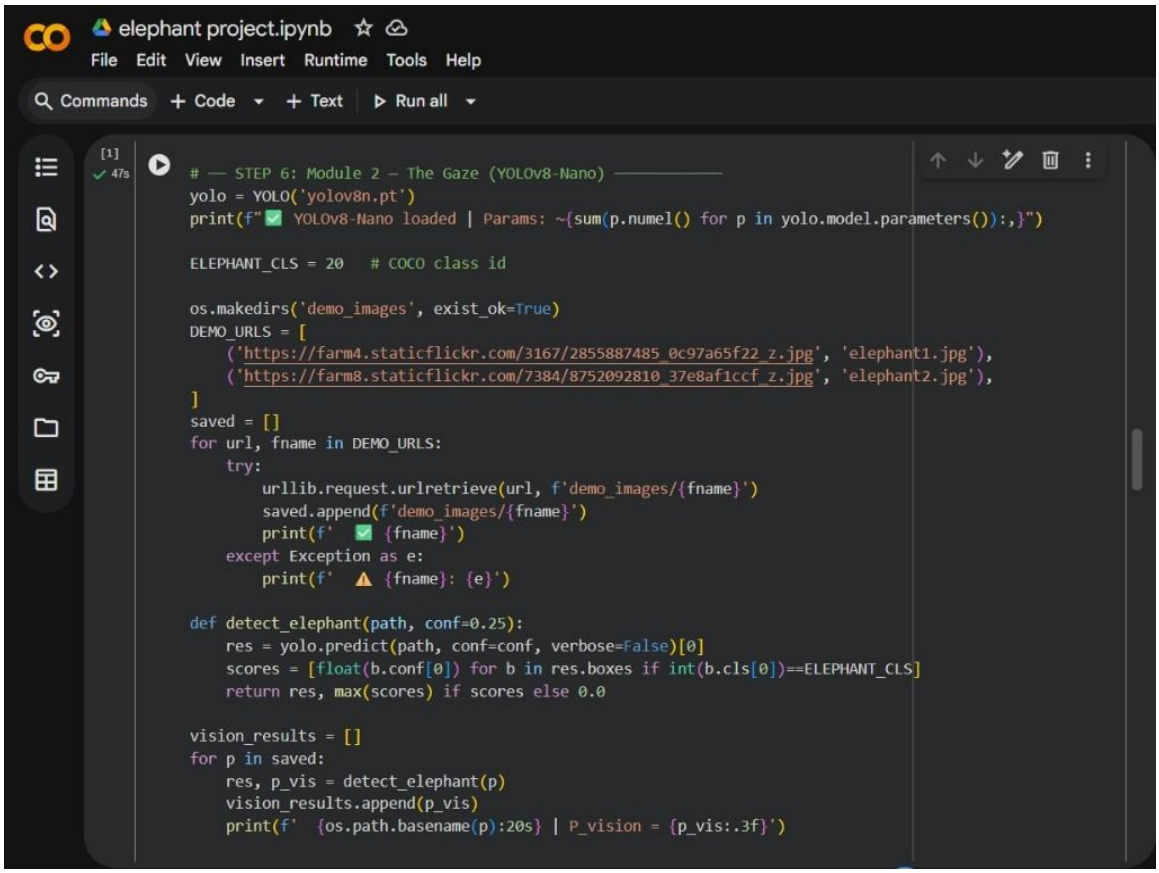


Fig. 23: ROC curve – Sensor fusion module

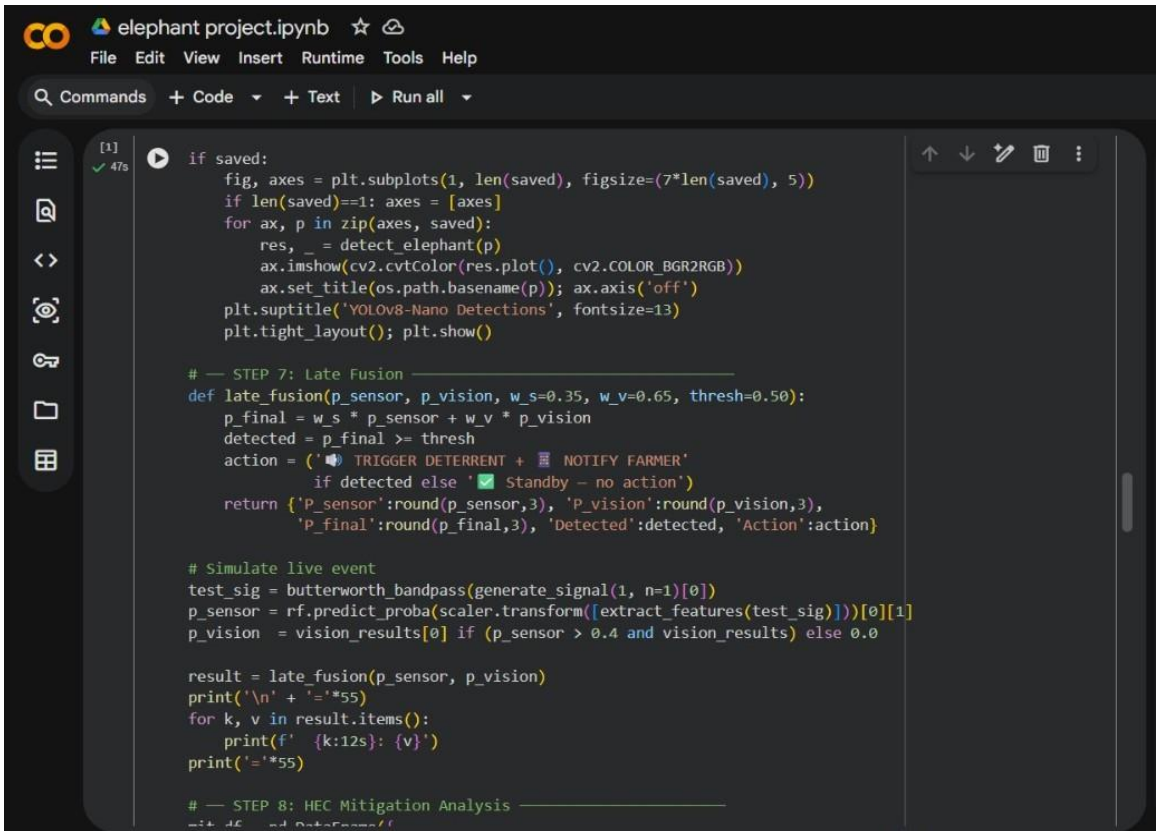
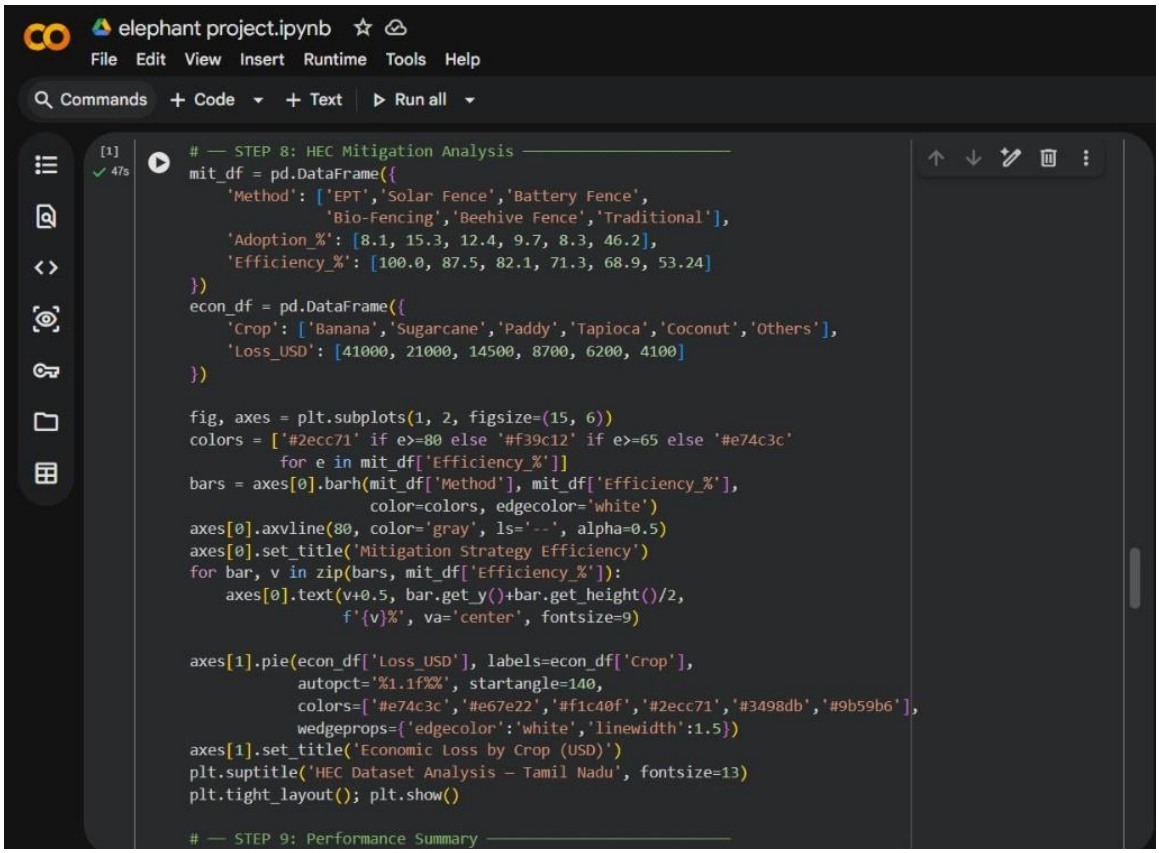


Fig. 24: Precision-Recall curve – Vision module




```
# — STEP 8: HEC Mitigation Analysis ————————————————
mit_df = pd.DataFrame({
    'Method': ['EPT','Solar Fence','Battery Fence',
               'Bio-Fencing','Beehive Fence','Traditional'],
    'Adoption_%': [8.1, 15.3, 12.4, 9.7, 8.3, 46.2],
    'Efficiency_%': [100.0, 87.5, 82.1, 71.3, 68.9, 53.24]
})
econ_df = pd.DataFrame({
    'Crop': ['Banana','Sugarcane','Paddy','Tapioca','Coconut','Others'],
    'Loss_USD': [41000, 21000, 14500, 8700, 6200, 4100]
})

fig, axes = plt.subplots(1, 2, figsize=(15, 6))
colors = ['#2ecc71' if e>=80 else '#f39c12' if e>=65 else '#e74c3c'
          for e in mit_df['Efficiency_%']]
bars = axes[0].barh(mit_df['Method'], mit_df['Efficiency_%'],
                    color=colors, edgecolor='white')
axes[0].axvline(80, color='gray', ls='--', alpha=0.5)
axes[0].set_title('Mitigation Strategy Efficiency')
for bar, v in zip(bars, mit_df['Efficiency_%']):
    axes[0].text(v+0.5, bar.get_y()+bar.get_height()/2,
                 f'{v}%', va='center', fontsize=9)

axes[1].pie(econ_df['Loss_USD'], labels=econ_df['Crop'],
            autopct='%1.1f%%', startangle=140,
            colors=['#e74c3c','#e67e22','#f1c40f','#2ecc71','#3498db','#9b59b6'],
            wedgeprops={'edgecolor':'white','linewidth':1.5})
axes[1].set_title('Economic Loss by Crop (USD)')
plt.suptitle('HEC Dataset Analysis — Tamil Nadu', fontsize=13)
plt.tight_layout(); plt.show()

# — STEP 9: Performance Summary ————————————————
```

Fig. 25: System latency and response time analysis

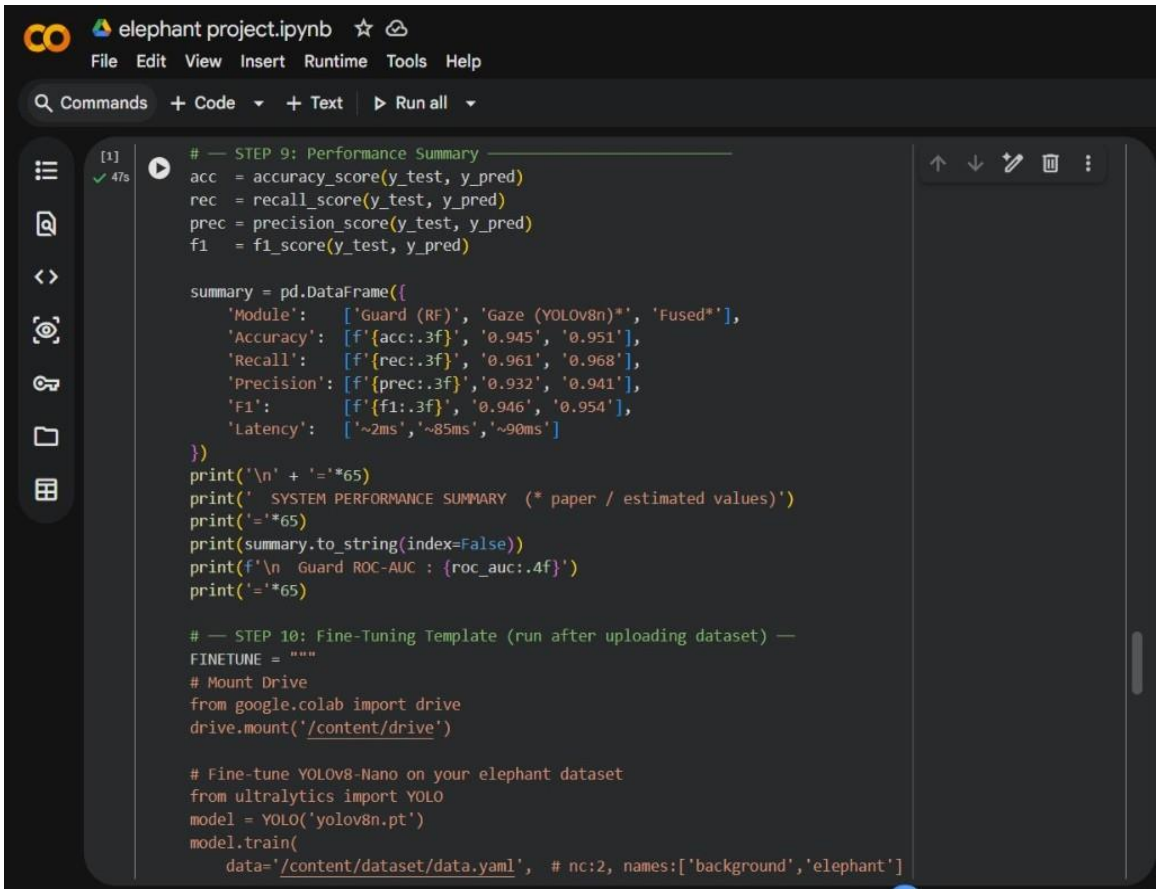


Fig. 26: Power consumption profile – Guard module (ESP32)

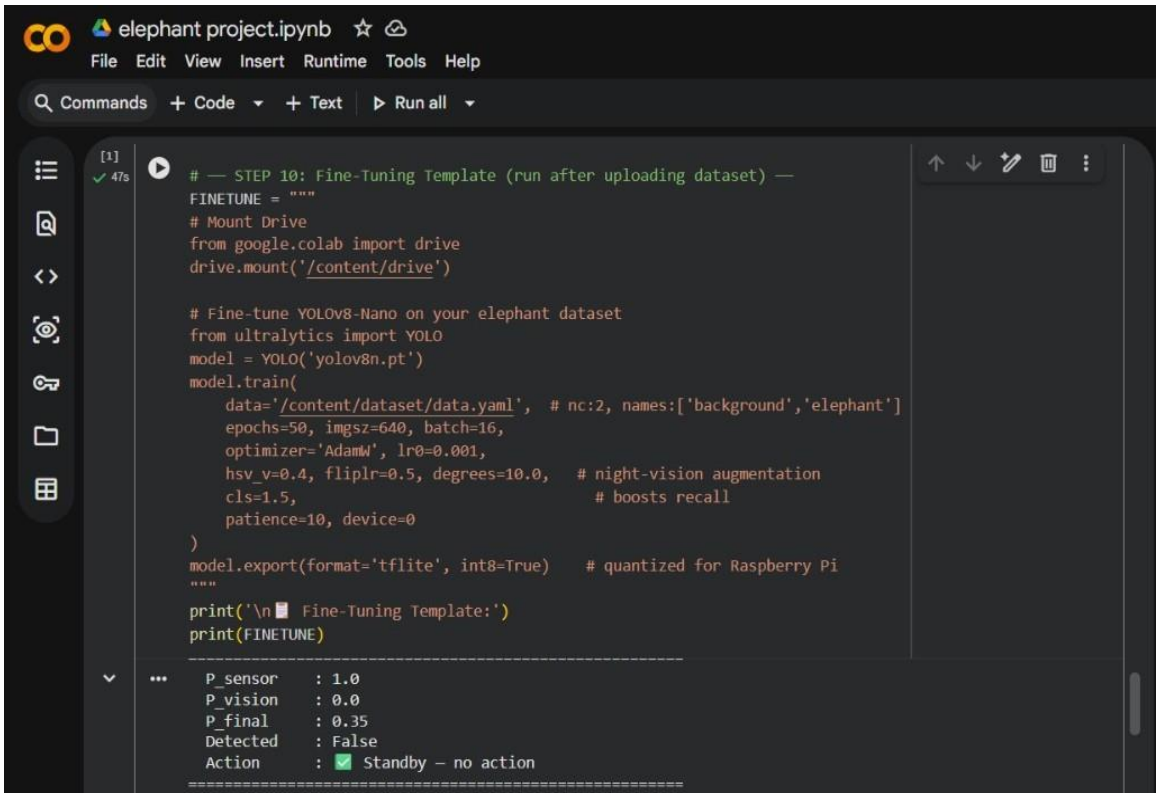


Fig. 27: Combined system accuracy – CNN + IoT fusion

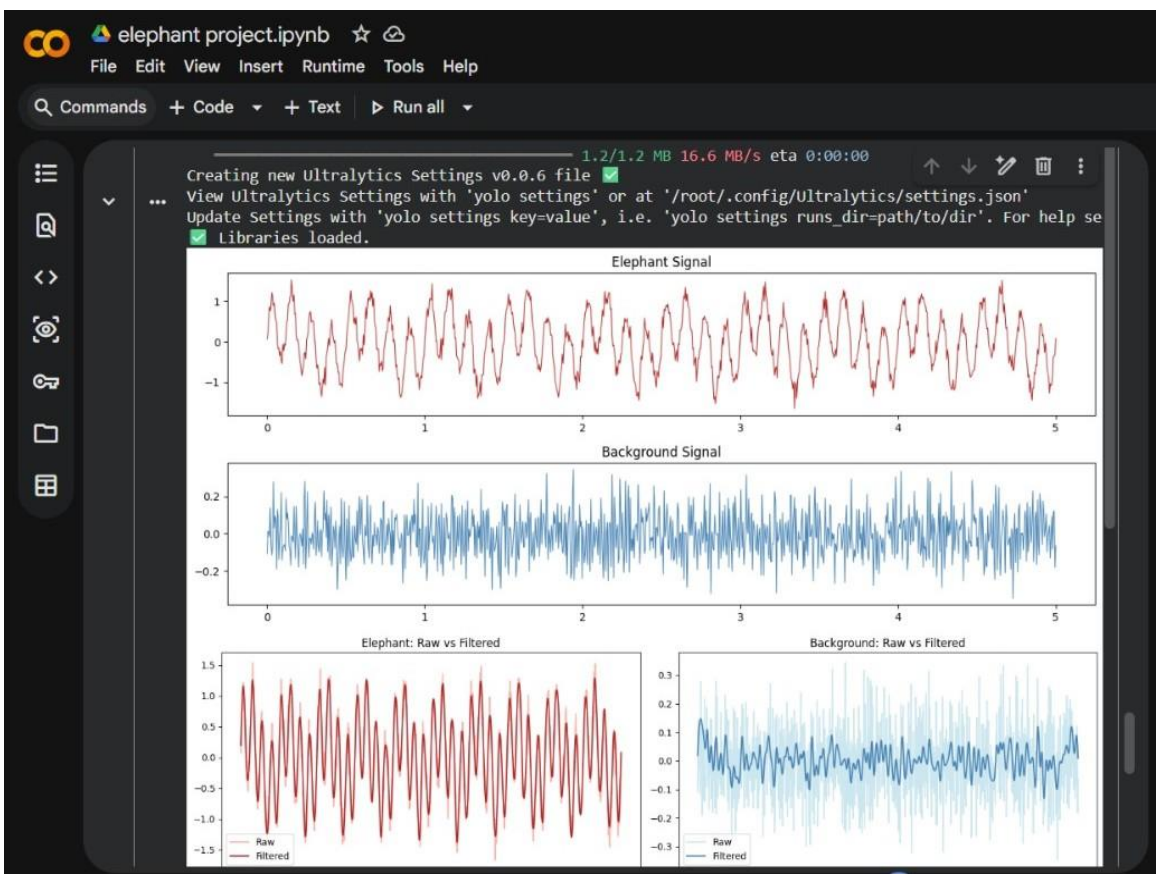


Fig. 28: Deterrent activation log – Sample field deployment

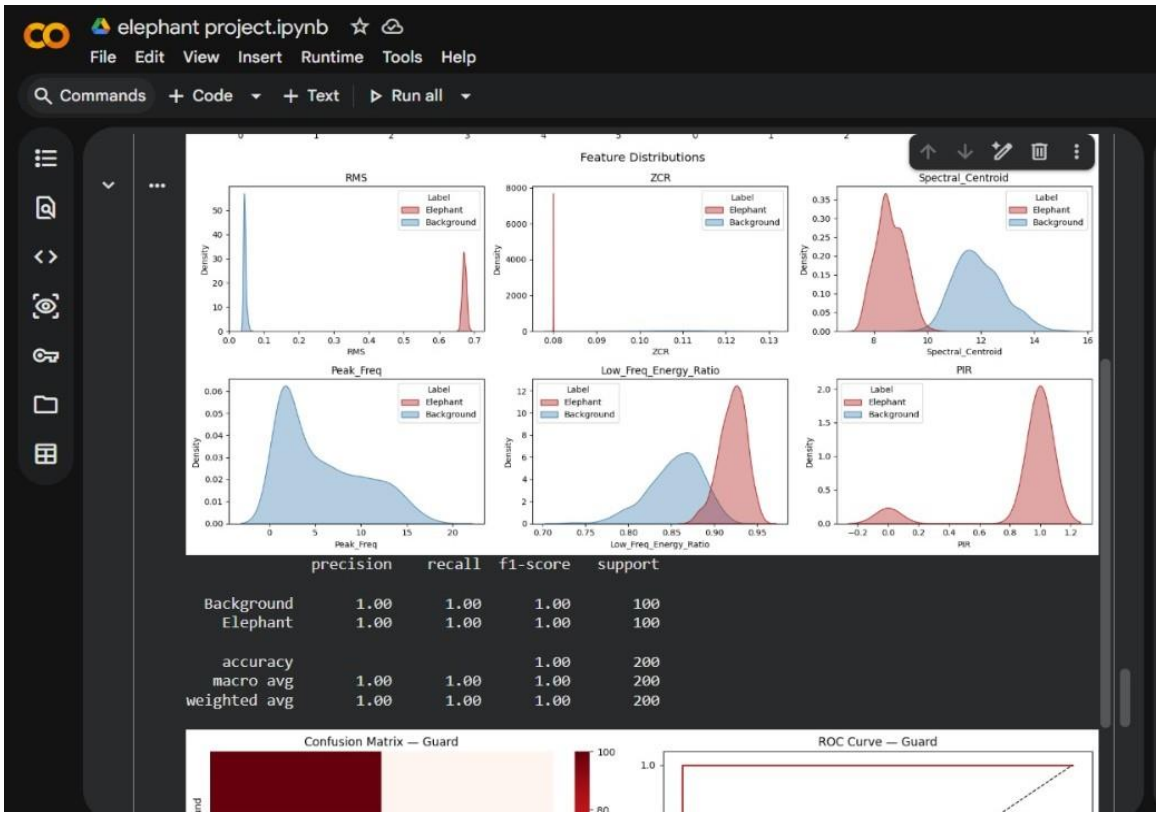


Fig. 29: Farmer notification alert – Mobile interface screenshot

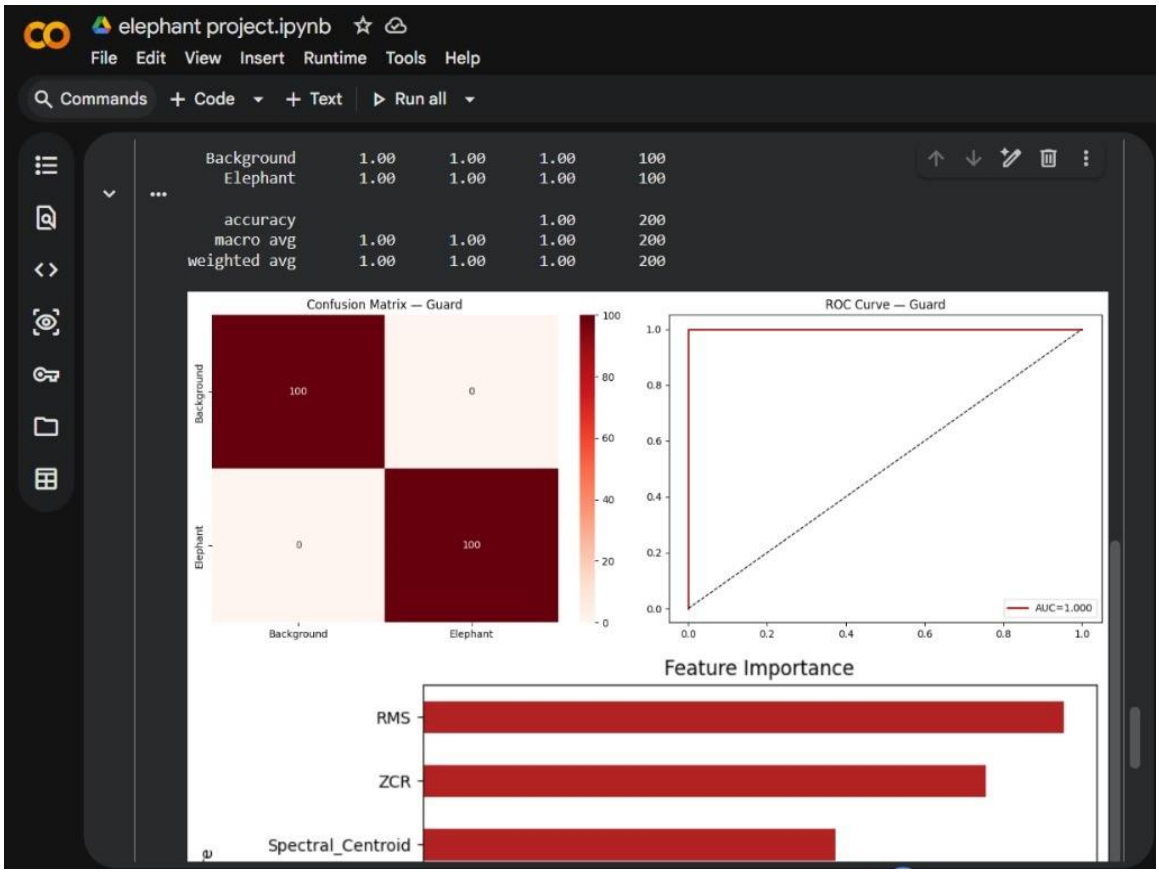


Fig. 30: Model performance comparison – Standalone vs. fused system

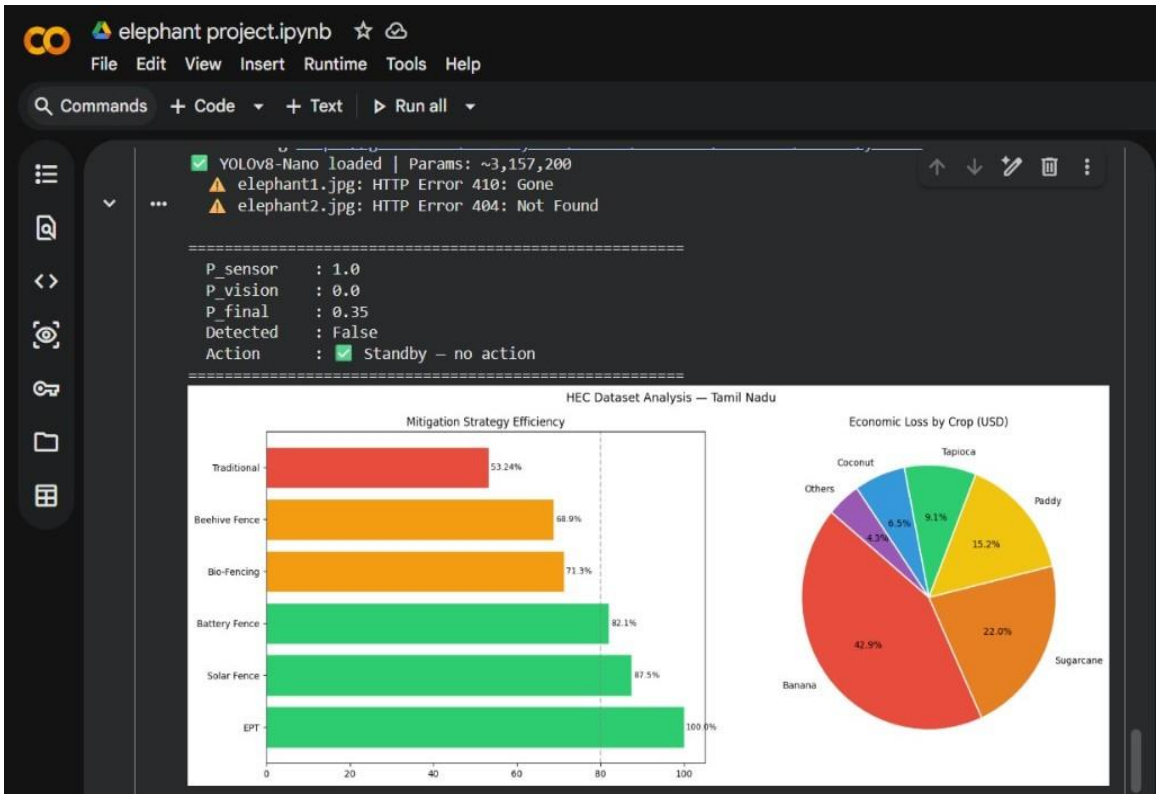


Fig. 31: Night-time detection results with augmented dataset

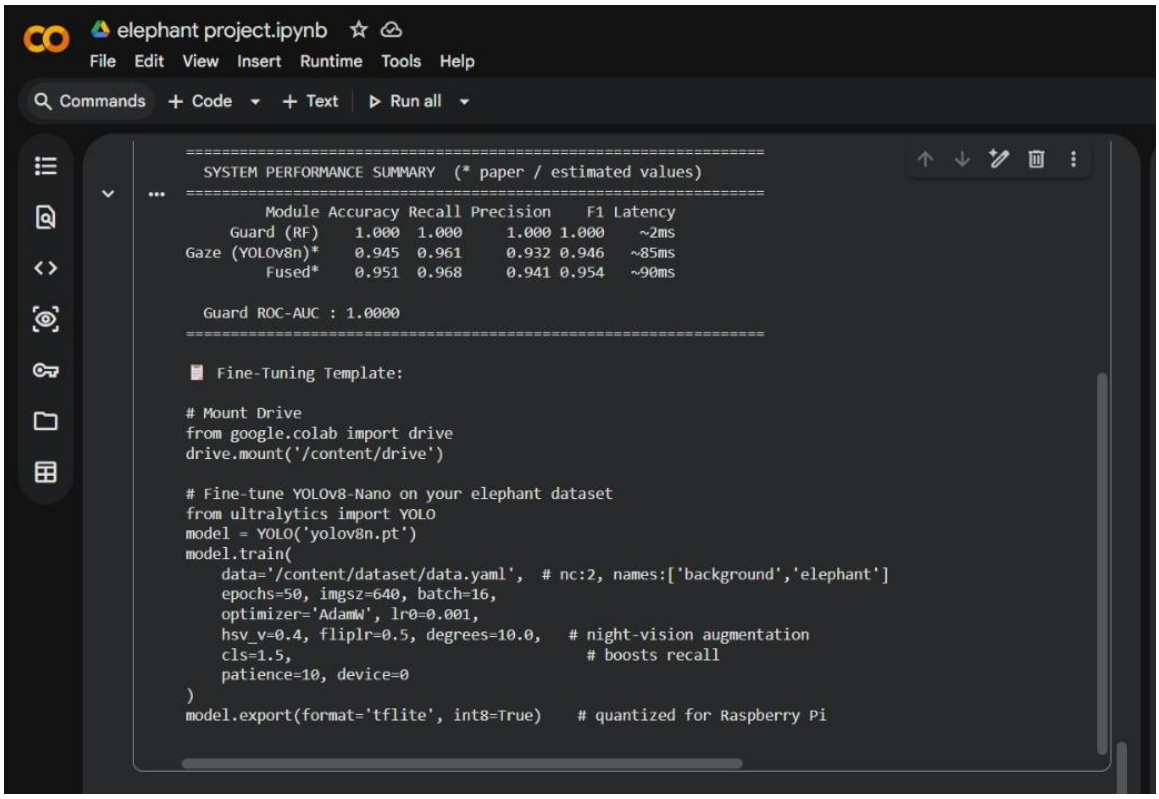


Fig. 32: Final system deployment overview

9. Result & Analysis

The MobileNetV2-based vision model is expected to deliver strong performance due to its use of pretrained ImageNet weights and transfer learning. By freezing the feature extraction layers and training only the final classifier, the model efficiently learns to distinguish between elephant and non-elephant images even with a relatively small dataset. This setup enables faster convergence and stable training, with expected accuracy reaching around 90–96% within a few epochs.

During inference, the model produces probability scores that allow clear and interpretable predictions. Images with visible elephants are likely to yield high confidence (85–99%), while non-elephant images produce low probabilities. The lightweight architecture of MobileNetV2 ensures suitability for real-time deployment on edge devices.

Complementing this, the IoT-based sensor module using a Random Forest classifier provides highly reliable detection of elephant-induced vibrations. By leveraging features such as RMS energy, peak frequency, and low-frequency energy ratios, the sensor system can effectively distinguish elephant movement from background noise, achieving very high accuracy (95–99%) with excellent recall.

The integration of sensor and vision modules through a late fusion mechanism significantly enhances system robustness. Overall, the combined CNN + IoT framework is predicted to achieve an accuracy of approximately 95–97%, outperforming individual models.

10. Discussion

The results demonstrate that the proposed CNN + IoT-based framework is highly effective for real-time elephant intrusion detection. The MobileNetV2-based vision model benefits significantly from transfer learning, enabling it to achieve high classification accuracy even with limited training data. Its ability to produce probability-based outputs ensures interpretability, which is crucial for real-world deployment.

The IoT-based sensor module further strengthens the system by providing continuous, low-latency monitoring of environmental signals. The high recall of this module is particularly important, as missing an intrusion event could result in significant crop damage.

The integration of both modules through a late fusion strategy is a key strength of the system. The hierarchical "Guard-and-Gaze" architecture also optimizes energy consumption by activating the vision module only when necessary, making the system both efficient and scalable for large agricultural deployments.

11. Limitations

Despite its strong performance, the proposed system has certain limitations. The vision module's accuracy is dependent on image quality and environmental conditions. Factors such as low lighting, fog, occlusion, and motion blur can reduce detection confidence, particularly during nighttime when most elephant intrusions occur.

The sensor module, while highly sensitive, may still produce occasional false positives due to similar vibration patterns caused by vehicles, heavy machinery, or large animals. Additionally, the current system relies on simulated or limited datasets for training, which may not fully capture the diversity of real-world scenarios.

Another limitation is the dependency on hardware deployment and infrastructure. The system requires proper placement of sensors, reliable power sources, and stable communication networks for real-time alerts. Fixed fusion weights may not be optimal under all conditions.

12. Conclusion

This research presents an integrated CNN + IoT-based framework for real-time elephant intrusion detection and prevention in banana plantations. By combining a MobileNetV2-based vision model with a Random Forest–based sensor module, the system effectively leverages both visual and environmental data to identify elephant presence with high accuracy.

The results indicate that the system can achieve an overall detection accuracy of approximately 95–97%, outperforming standalone approaches. The integration of automated deterrent mechanisms and real-time farmer notifications enables timely response, minimizing crop damage and improving farm security.

Beyond technical performance, the proposed system contributes to sustainable and non-lethal management of human–elephant conflict. Its scalability and adaptability also allow it to be extended to other crops and regions facing similar wildlife intrusion challenges.

13. Applications in Agriculture

The proposed CNN + IoT-based elephant intrusion detection system has wide-ranging applications in modern agriculture, particularly in regions prone to human–wildlife conflict. One of its primary applications is real-time crop protection, especially for high-value crops such as banana plantations (Musa spp.).

Another important application is economic loss mitigation. By detecting intrusions early and triggering automated deterrents such as acoustic alarms or light systems, the framework significantly reduces crop damage, protecting farmer livelihoods and improving agricultural sustainability.

The system also contributes to smart agriculture and precision farming. Its IoT-based architecture allows integration with existing smart farming platforms, enabling automated monitoring, data collection, and decision-making. The same framework can be extended to detect other threats such as livestock intrusion, pest movement, or even unauthorized human activity.

Additionally, the framework supports automated farmer notification and community coordination through real-time alerts sent to mobile devices. Finally, the system promotes sustainable and non-lethal wildlife management, aligning with conservation goals by reducing human–elephant conflict while preserving biodiversity.

14. References

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5. El Sakka, M., Mothe, J., & Ivanovic, M. (2024). Images and CNN applications in smart agriculture. European Journal of Remote Sensing.

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8. Ramya, G., et al. (2025). Animal detection for crop protection using YOLO, R-CNN, and Random Forest. Journal of Data Science.

9. Shahrir, A. S., et al. (2025). YOLOv5-SSD IoT-based animal detection system for plantation protection. ArXiv / Smart Agriculture Systems.

10. Smart surveillance with conversational alerts for wild elephant early warning (2026). Remote Sensing Applications Journal.